

The Power of Power Law:

Asymmetry Enables Compositional Reasoning

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Where should we shift the training data?

A common view: the power-law learning curve is induced by an underlying power-law distribution over latent “skills” / “knowledge pieces” in natural data.

(Michaud et al. 2023; Arora & Goyal 2023)

CONCEPTUAL QUESTION

If we had infinite compute for data curation,
should we always shift the training data toward UNIFORM?

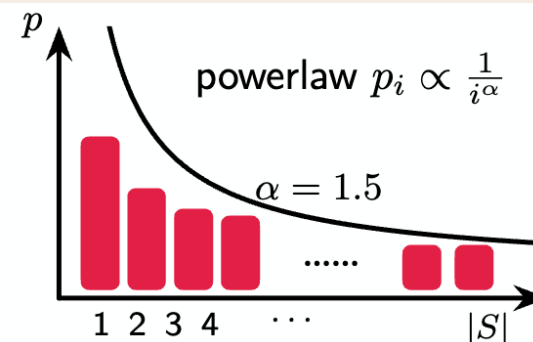
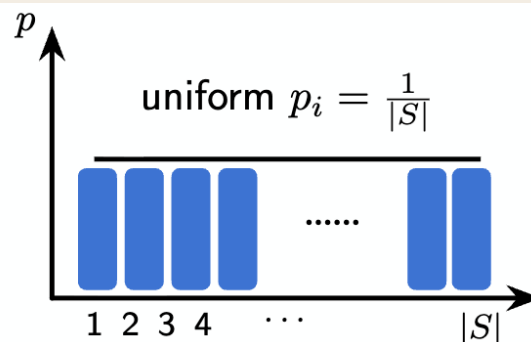
Our answer: NO.

Even when skills are equally hard, asymmetric (Zipf) training **BEATS** uniform
Even the test set is uniform

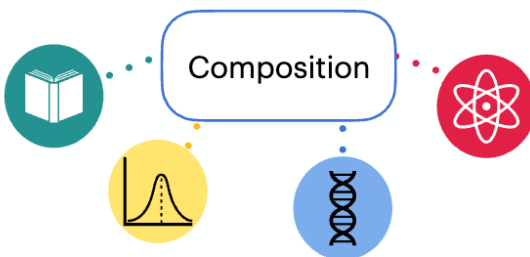
Two synthetic compositional tasks

Skill distribution: $i \sim p_i$

$a \in [n]$ or $g_i \in S_n$



Compositional tasks



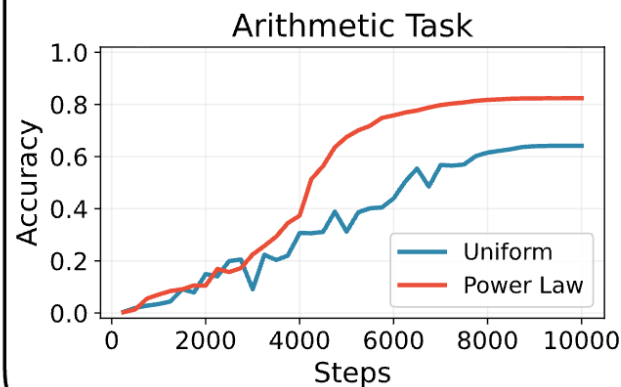
Test on uniform

— uniform 👎

— power law 👍

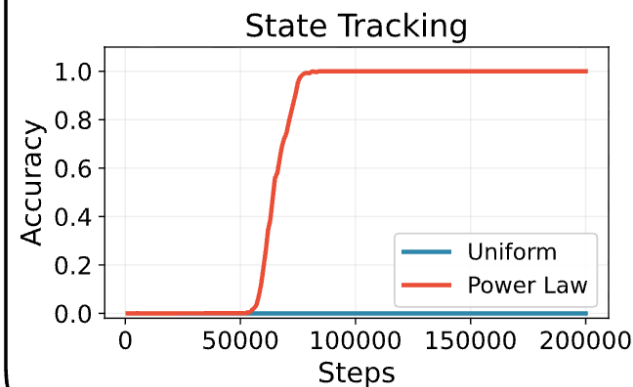
Multi-step arithmetic

$a \times b + c - d = ?$



State tracking

$g_1 \circ g_2 \circ g_3 \circ g_4 = ?$



Power-law (Zipf, $\alpha=1.5$) consistently beats uniform — both on multi-step arithmetic and S_5 state tracking.

A minimalist abstraction for skill composition.

Input: A sequence of skill IDs, $\mathbf{X} = (s_1, \dots, s_k), s_i \in [d]$

Ground Truth Skill Vector: $\mathbf{w}^* \in \{-1, +1\}^d$

Target: $y(\mathbf{X}) = \prod_{i=1}^k w_{s_i}^*$

Model:

- Embed each s_i to $\mathbf{x}_i = \mathbf{e}_{s_i} \in \mathbb{R}^d$
- Compute $f(\mathbf{X}; \mathbf{w}) = \prod_{i=1}^k \langle \mathbf{w}, \mathbf{x}_i \rangle$ (can be seen as an RNN)

Connection to Function Composition:

- Each skill is a multiplication function $g_s(h) = w_s^* \cdot h$
- The task is essentially asking for $g_{s_1} \circ g_{s_2} \circ \dots \circ g_{s_k}$

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Question: If you train the model on n samples from D ...

- **Choice 1: Uniform** $D: s_i \sim \text{uniform}(\{1, \dots, d\})$
- **Choice 2: Power Law** $\Pr_D[s_i = j] \propto j^{-a}$

Which one will lead to a smaller test loss on the **uniform** distribution?

Under uniform, no efficient gradient learning algorithm.

Theorem 1 *(CSQ Lower Bound for Uniform).*

Under uniform $p_j = 1/d$ with $k \geq 2$, achieving test loss $< O(1)$ requires either

$$n \gtrsim \tilde{\Omega}(d^{k/2}) \text{ samples} \quad \text{or runtime } \exp(d).$$

Symmetry of the function class:

gradients cannot distinguish w^ from a sign-flipped neighbor.*

Asymmetry breaks the lower bound.

Theorem 2 (*Online SGD on Power Law*).

Under Zipf $p_j \propto j^{-\alpha}$ with $\alpha > 1$, online minibatch SGD with proper η , B reaches error ε with

$$n \leq \tilde{O}\left(\frac{d^{2\alpha}}{\varepsilon}\right) \text{ samples, } t \leq \tilde{O}(d^\alpha \log(1/\varepsilon)) \text{ iterations.}$$

	Uniform	Zipf ($\alpha = 1$)
Samples	$d^{k/2}$	d^2
Compute	$d^{k/2}$	$d \log(1/\varepsilon)$

Zipf upper bound is polynomial in d for a fixed k

Power law creates its own curriculum through optimization

I · ESCAPE

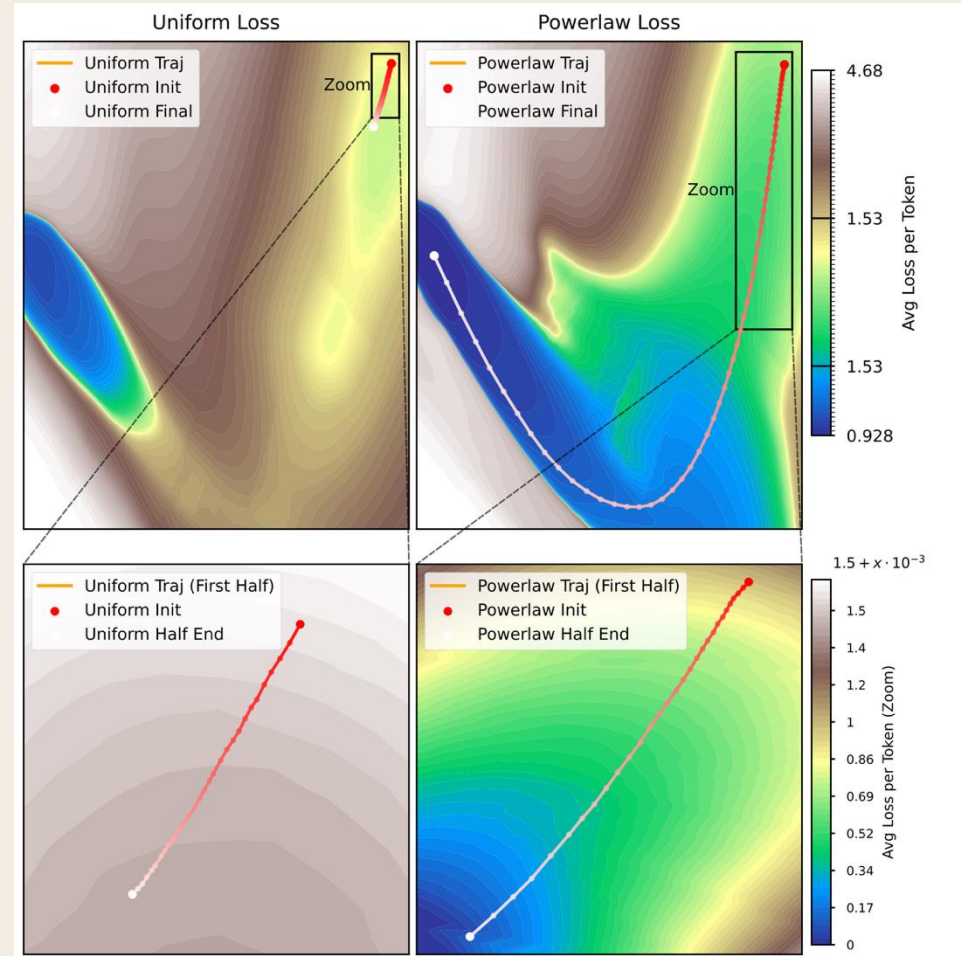
Head skills get a real gradient early;
uniform training stays trapped in a flat basin.

II · BOOTSTRAP

head skills align first. $A(t)$ grows, amplifying tail-skill gradients through $A(t)^{k-1}$.

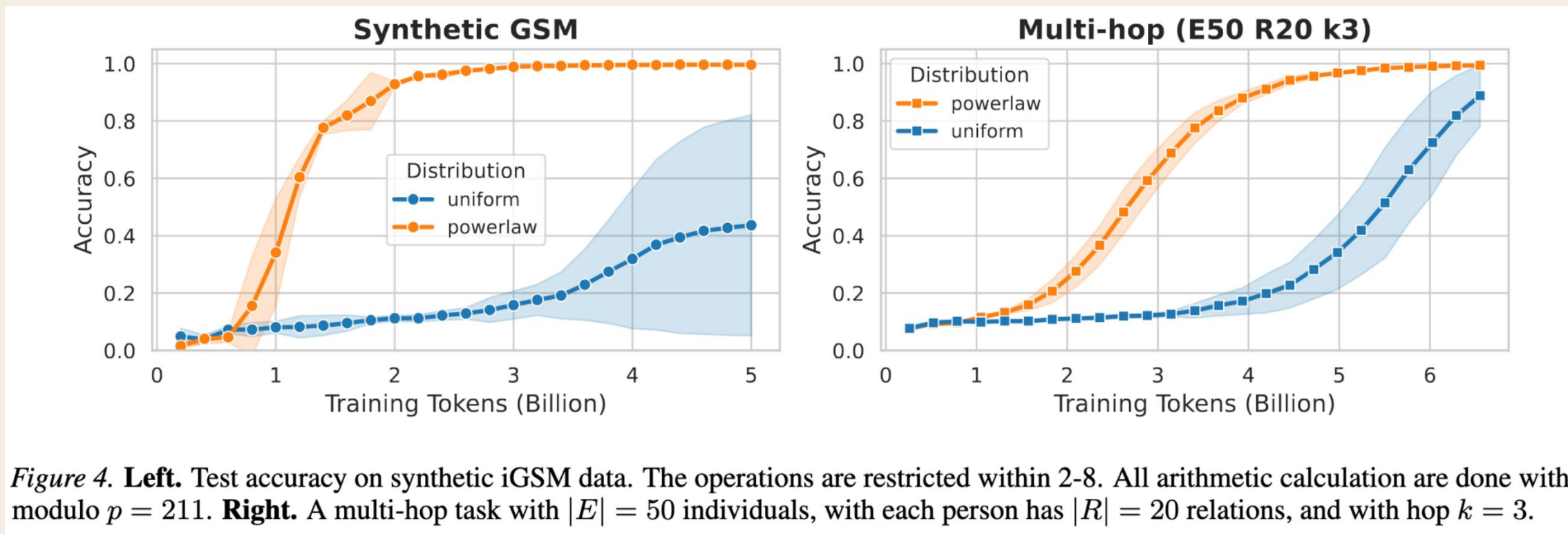
III · TAIL

rare skills converge last; composition has already emerged.



Loss landscape at init. Uniform stays flat; power-law has a steep descent channel.

Holds on natural-language reasoning tasks too.



Synthetic iGSM (Ye et al. 2024) and multi-hop QA (Yao et al. 2025).

Empirical validation focuses on synthetic tasks — there we know what the latent skills are.

Skew the skills, not just the examples.

Uniformity is not always neutral

it can preserve the symmetry that makes composition hard to learn.

→ REAL-WORLD SKILL COMPOSITIONS

Agentic workflows, tool-use traces — power-law structure already exists. Can we induce it during pretraining or skill-mix construction?

→ GRANULARITY OF ASYMMETRY

Power-law is one of many asymmetric distributions. Ablations suggest finer granularity helps more — what's the right design space?